

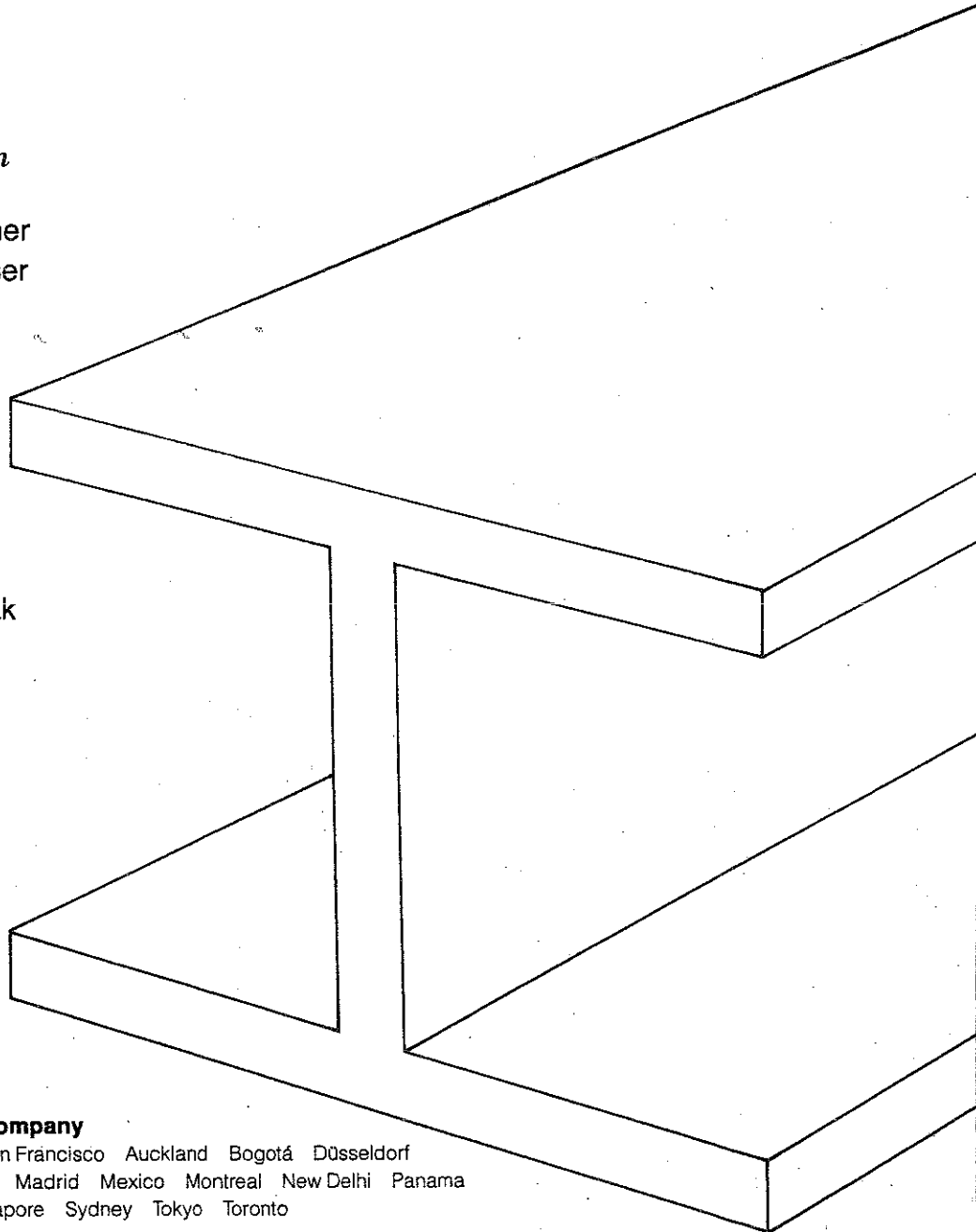
Solutions Manual to accompany
Crandall/Dahl/Lardner:
**AN INTRODUCTION TO
THE MECHANICS
OF SOLIDS**
SECOND EDITION WITH SI UNITS

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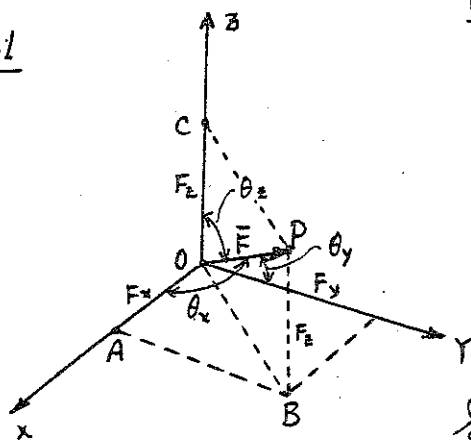
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CHAPTER 1

1.1



$CP \parallel OB$, $OB \perp OC$

$\therefore CP \perp OC$

It follows then, that

$$\cos \theta_z = \frac{F_z}{|F|}$$

Similarly $\cos \theta_y = \frac{F_y}{|F|}$

$$\cos \theta_x = \frac{F_x}{|F|}$$

By Pythagoras' Theorem, $F_x^2 + F_y^2 + F_z^2 = |F|^2$

i.e., $|F|^2 \cos^2 \theta_x + |F|^2 \cos^2 \theta_y + |F|^2 \cos^2 \theta_z = |F|^2$, or $\cos^2 \theta_x + \cos^2 \theta_y + \cos^2 \theta_z = 1$

1.2 See from figure that

$$\bar{i} = \bar{a} \cos \theta - \bar{b} \sin \theta$$

$$\bar{j} = \bar{a} \sin \theta + \bar{b} \cos \theta$$

We also have $\bar{i} \cdot \bar{a} = |\bar{i}| |\bar{a}| \cos \theta = \cos \theta$

$$\bar{i} \cdot \bar{b} = -\sin \theta$$

$$\bar{j} \cdot \bar{a} = \cos(\pi/2 - \theta) = \sin \theta$$

$$\bar{j} \cdot \bar{b} = \cos \theta$$

$\therefore$ More generally, $\bar{i} = (\bar{i} \cdot \bar{a}) \bar{a} + (\bar{i} \cdot \bar{b}) \bar{b}$

$$\bar{j} = (\bar{j} \cdot \bar{a}) \bar{a} + (\bar{j} \cdot \bar{b}) \bar{b}$$

Then $\bar{F} = F_x \bar{i} + F_y \bar{j} = F_x (\bar{a} \cos \theta - \bar{b} \sin \theta) + F_y (\bar{a} \sin \theta + \bar{b} \cos \theta)$

$$= \bar{a} (F_x \cos \theta + F_y \sin \theta) + \bar{b} (F_y \cos \theta - F_x \sin \theta)$$

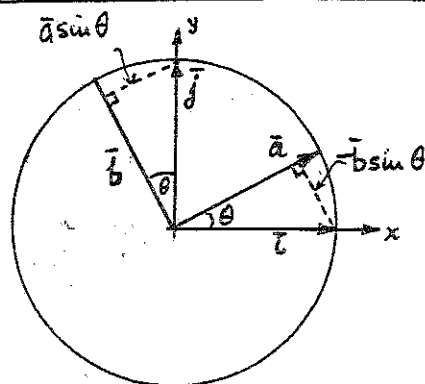
$\therefore F_a = F_x \cos \theta + F_y \sin \theta$ ----- (i)

and $F_b = -F_x \sin \theta + F_y \cos \theta$ ----- (ii)

From (i) and (ii), we can get $F_y = F_a \sin \theta + F_b \cos \theta$

$$F_x = F_a \cos \theta - F_b \sin \theta$$

MATRIX FORM FOLLOWS IMMEDIATELY



1.3 We have $\vec{r} = \vec{i}x + \vec{j}y$

$$\vec{F}_1 = \vec{i}F_{x1} + \vec{j}F_{y1}$$

$$\vec{F}_2 = \vec{i}F_{x2} + \vec{j}F_{y2}$$

Then $\vec{M}_0 = \vec{r} \times (\vec{F}_1 + \vec{F}_2)$

$$= (\vec{i}x + \vec{j}y) \times [\vec{i}(F_{x1} + F_{x2}) + \vec{j}(F_{y1} + F_{y2})]$$

$$= \{x(F_{y1} + F_{y2}) - y(F_{x1} + F_{x2})\} \vec{k}$$

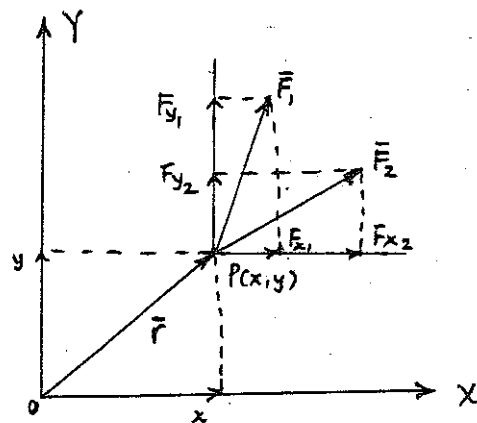
Now consider moments of $\vec{F}_1$ and $\vec{F}_2$ separately:

$$\vec{M}_{01} = \vec{r} \times \vec{F}_1 = (xF_{y1} - yF_{x1}) \vec{k}$$

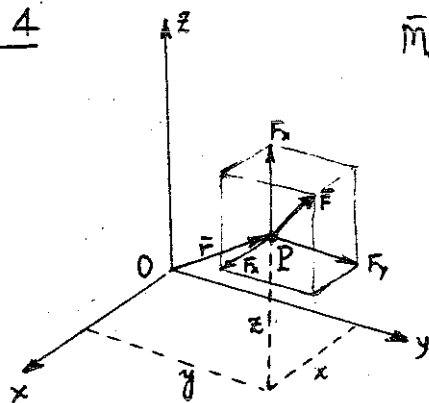
$$\vec{M}_{02} = \vec{r} \times \vec{F}_2 = (xF_{y2} - yF_{x2}) \vec{k}$$

and $\sum_{i=1}^2 \vec{M}_{0i} = \{x(F_{y1} + F_{y2}) - y(F_{x1} + F_{x2})\} \vec{k} = \vec{M}_0$

This shows that $\vec{r} \times \vec{F}_1 + \vec{r} \times \vec{F}_2 = \vec{r} \times (\vec{F}_1 + \vec{F}_2)$



1.4



$$\vec{M}_0 = \text{Moment about } O = \vec{r} \times \vec{F}$$

$$= (x\vec{i} + y\vec{j} + z\vec{k}) \times (F_x\vec{i} + F_y\vec{j} + F_z\vec{k})$$

$$= \vec{i}(yF_z - zF_y) + \vec{j}(zF_x - xF_z) + \vec{k}(xF_y - yF_x)$$

which is the value of $\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ x & y & z \\ F_x & F_y & F_z \end{vmatrix}$

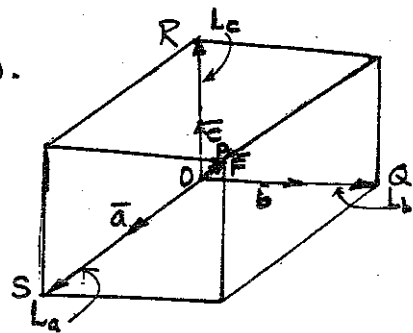
1.5 $\vec{OP} \equiv \vec{F}$.

Draw planes through P parallel to $(\vec{a}, \vec{b})$, $(\vec{b}, \vec{c})$, $(\vec{c}, \vec{a})$.

Let these intersect $\vec{c}, \vec{a}, \vec{b}$ in R, S and Q.

Then $OP = L_a$; $OQ = L_b$; $OR = L_c$.

Thus the components are obtained by forming a parallelepiped with $\vec{F}$ as a main diagonal and $\vec{a}, \vec{b}, \vec{c}$ lying along three contiguous edges.



continued

1.5 continued

$$(\bar{F} \times \bar{b}) \cdot \bar{c} = \{ (l_a \bar{a} + l_b \bar{b} + l_c \bar{c}) \times \bar{b} \} \cdot \bar{c}$$

$$= l_a (\bar{a} \times \bar{b}) \cdot \bar{c} + l_b (\bar{b} \times \bar{b}) \cdot \bar{c} + l_c (\bar{c} \times \bar{b}) \cdot \bar{c}$$

Now $(\bar{c} \times \bar{b}) \cdot \bar{c} = 0$ because $\bar{c} \times \bar{b} \perp \bar{c}$

also $(\bar{b} \times \bar{b}) = 0$ by definition of vector product

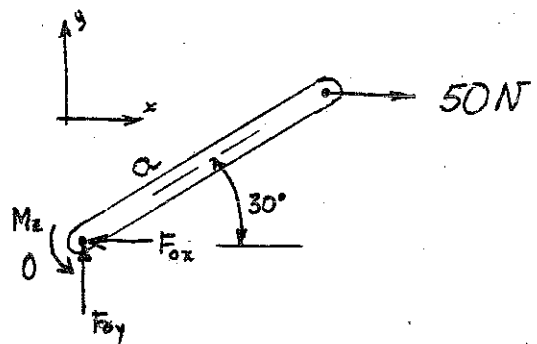
$$\therefore (\bar{F} \times \bar{b}) \cdot \bar{c} = l_a (\bar{a} \times \bar{b}) \cdot \bar{c} \quad \therefore l_a = \frac{(\bar{F} \times \bar{b}) \cdot \bar{c}}{(\bar{a} \times \bar{b}) \cdot \bar{c}}$$

Similar expressions are easily obtained for l_b and l_c ,

$$l_b = \frac{(\bar{F} \times \bar{c}) \cdot \bar{a}}{(\bar{a} \times \bar{b}) \cdot \bar{c}} \quad \text{and} \quad l_c = \frac{(\bar{F} \times \bar{a}) \cdot \bar{b}}{(\bar{a} \times \bar{b}) \cdot \bar{c}}$$

1.6 Consider the free-body shown.

The forces at O are those exerted on the bar by the support. These can at most be forces along two perpendicular directions x and y and a moment vector M_z along the z -axis, indicated by the couple shown.



Equilibrium: $\sum F_x = 0 : F_{Ox} = 50 \text{ N}$

$$\sum F_y = 0 : F_{Oy} = 0$$

$$\sum M_{Oz} = 0 : M_z = 50 \times 0.3 \sin 30^\circ = 7.5 \text{ N}\cdot\text{m}$$

1.7 (a) $\sum F_a = 0$, $\sum F_b = 0$ assures $\sum \bar{F} = 0$, but allows the possibility of a resultant couple in the x - y plane. The moment of this couple about any point is the same. Therefore, if one sets $\sum M_o = 0$ about any point in the x - y plane, the possibility of a couple is ruled out.

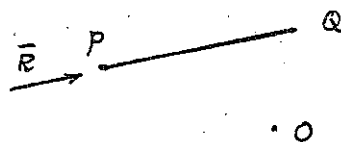
(b) $\sum M_P = 0$
 $\sum M_Q = 0$ } Rules out possibility of a couple but allows a resultant force $\bar{R}$ acting along PQ



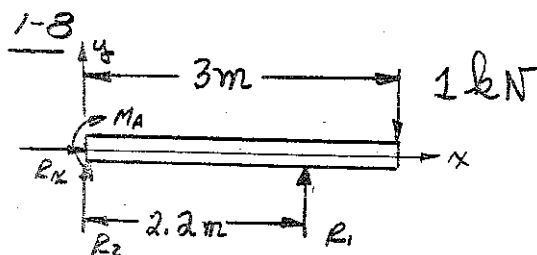
1.7 continued

Setting $\Sigma \vec{F}_{PQ} = 0$ in addition ensures that there is no resultant force.

- (c) $\Sigma M_P = 0$, $\Sigma M_Q = 0$ rules out possibility of a couple, but allows resultant force along PQ.



$\Sigma M_O = 0$ rules out possibility of force resultant along PQ.



$$\Sigma F_x = 0$$

$$R_x = 0$$

$$\Sigma F_y = 0$$

$$-1000 + R_1 + R_2 = 0$$

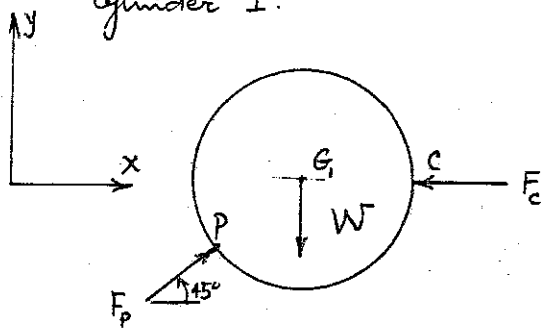
$$+\circlearrowleft \Sigma M_O = 0$$

$$-M_A + 2.2R_1 - 3 \times 1000 = 0$$

THE STRUCTURE IS STATICALLY INDETERMINANT. THERE ARE TWO INDEPENDENT EQUILIBRIUM EQUATIONS FOR THE THREE UNKNOWN M_A , R_1 , R_2 . METHODS OF SOLUTION FOR THIS TYPE OF PROBLEM ARE OUTLINED IN CHAPTER 8.

1.9 Idealisation: Assume cylinders and box to be rigid: reactions are all point forces.

Equilibrium: consider a free-body of cylinder I.



$$W = 900 \text{ N}$$

$\Sigma M_{G_1} = 0$: Satisfied if there is no friction at P & C, and F_p, F_c radial.

$$\Sigma F_y = 0: F_p \sin 45^\circ = W \therefore F_p = \frac{2W}{\sqrt{2}}$$

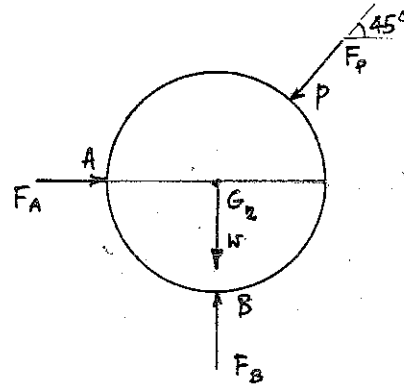
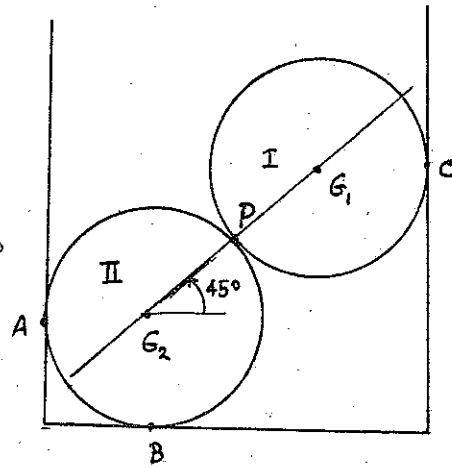
$$\Sigma F_x = 0: F_p \cos 45^\circ = F_c \therefore F_c = W$$

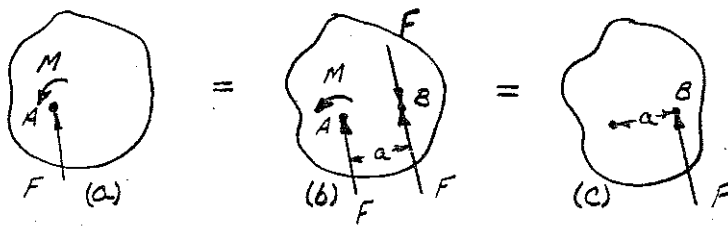
Free-body of cylinder II:

$$\Sigma F_x = 0: F_A = F_p \cos 45^\circ = W$$

$$\Sigma F_y = 0: F_B = W + F_p \sin 45^\circ = 2W$$

$$\therefore \left. \begin{array}{l} F_A = 900 \text{ N} \\ F_B = 1800 \text{ N} \\ F_c = 900 \text{ N} \end{array} \right\} \text{ANSWERS.}$$





THE ADDITION OF A PAIR OF EQUAL AND OPPOSITE FORCES DOES NOT ALTER THE REQUIRED FORCE TO PLACE THE BODY IN EQUILIBRIUM. THE ORIGINAL FORCE AND THE ADDED FORCE OF OPPOSITE DIRECTION FORM A COUPLE ABOUT A OF MAGNITUDE $|F|a$. THE SENSE OF THIS COUPLE IS OPPOSITE THAT OF THE APPLIED MOMENT M . SO THAT THE TWO MOMENTS WILL ADD TO ZERO IF

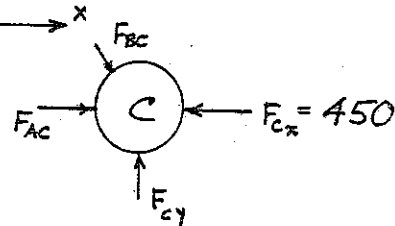
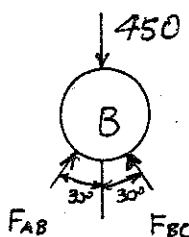
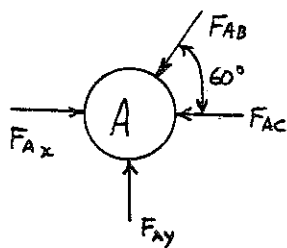
$$|F|a = |M| \text{ OR } a = |M| / |F|$$

THE REMAINING FORCE AT B WILL HAVE A MOMENT OF EQUAL MAGNITUDE AND SIMILAR SENSE AS THE APPLIED MOMENT. THE FORCES REQUIRED FOR EQUILIBRIUM ARE THE SAME AS IN (a)

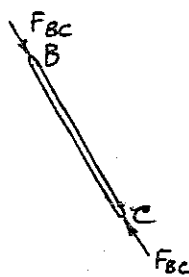
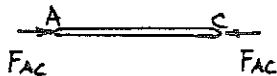
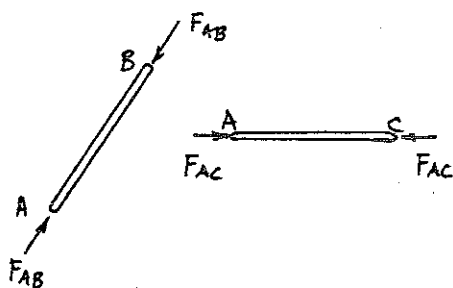
1-11

Idealisation: bars are 2-force members (weightless), and therefore forces in bars are along their axes. No friction at pins. Negligible deformation.

Free-bodies of pins:



Free-bodies of bars:



Equilibrium of pin B:

$$\sum F_x = 0:$$

$$F_{BC} \sin 30^\circ = F_{AB} \sin 30^\circ$$

$$\therefore F_{BC} = F_{AB}$$

$$\sum F_y = 0$$

$$(F_{BC} + F_{AB}) \cos 30^\circ = 450$$

$$\therefore \underline{F_{AB} = F_{BC} = 260 \text{ N}}$$

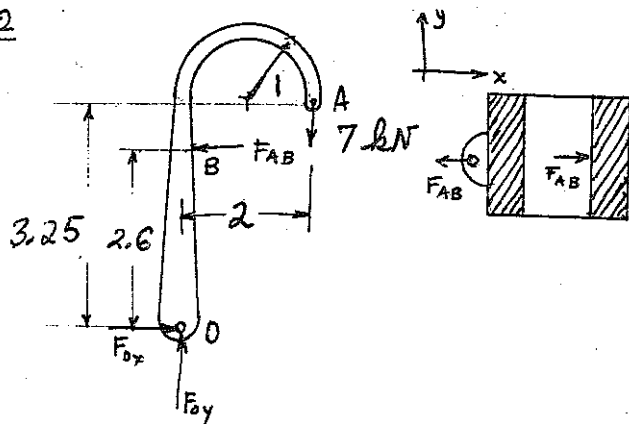
Equilibrium of pin C:

$$\sum F_x = 0: F_{AC} - F_{BC} \cos 60^\circ = 0$$

$$\therefore \underline{F_{AC} = 320 \text{ N}}$$

NOTE: ALL BARS ARE IN COMPRESSION

1.12



Idealisation: No friction at A, B, O.

All forces are in plane of paper

Equilibrium: $\sum M_O = 0$:

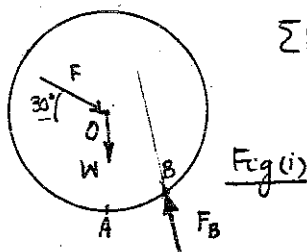
$$7 \times 2 - F_{AB} \times 2.6 = 0$$

$$\therefore \underline{F_{AB} = 5.38 \text{ kN}}$$

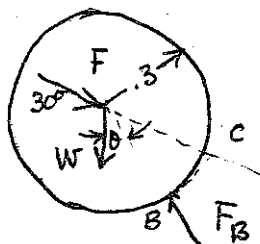
1.13 Idealisation: Inelastic contact. No friction at A. No slip at B.

At instant when roller begins to rise, force at A = 0. If F is just enough to cause the roller to rise, it is still in static equilibrium.

(a) Pushing.



$\sum M_O = 0$. This requires F_B to pass through O. Thus the free-body appears as in Fig (ii)



$$a = 0.3 \text{ m}$$

EQUILIBRIUM: $\sum M_B = 0 \Rightarrow$

$$W a \sin \theta = F \times BC = F a \sin (60 - \theta)$$

$$F = \frac{W \sin \theta}{\sin (60 - \theta)} = 1866 \text{ N}$$

$$\cos \theta = \frac{300 - 75}{300} \quad \therefore \theta = 41.4^\circ$$

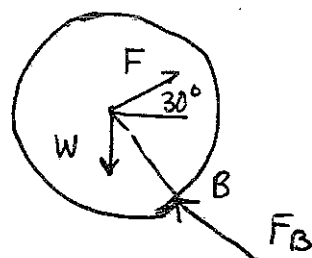
(b) PULLING

$$\sum M_B = 0 \quad W a \sin \theta = F a \sin (90 - \theta + 30)$$

$$F = 607 \text{ N}$$

PUSHING $F = 1866 \text{ N}$

PULLING $F = 607 \text{ N}$



1.14 Idealisation: assume that bearings at A and B can transmit no moments

Equilibrium of free-body:

$$\sum F_x = 0 : F_{Ax} + F_{Bx} = 0 \quad \text{--- (i)}$$

$$\sum F_y = 0 : F_{Ay} + F_{By} = 0 \quad \text{--- (ii)}$$

$$\sum F_z = 0 : F_{Az} + F_{Bz} = 900 \text{ N} \quad \text{--- (iii)}$$

From configuration, $F_{Az} = 0$ $\therefore F_{Bz} = 900 \text{ N}$

$$\sum M_{Bx} = 0 \quad -900(225) - F_{Ay}(150) = 0$$

$$\sum M_{By} = 0 \quad 900(300) + F_{Ax}(150) = 0$$

$$\sum M_{Bz} = 0 \quad \text{IDENTICALLY SATISFIED}$$

SOLVING

$$F_{Bz} = 900 \text{ N}$$

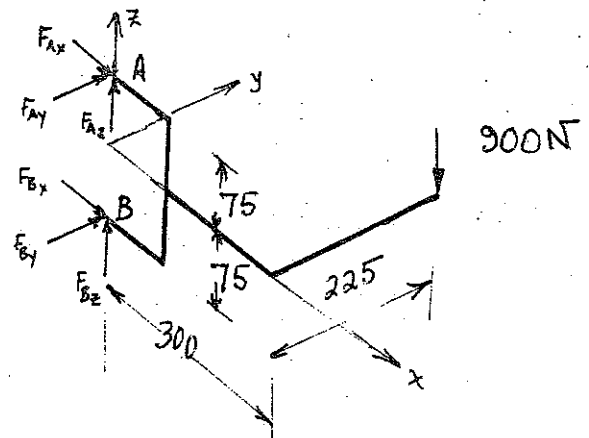
$$F_{Az} = 0$$

$$F_{Bx} = 1800 \text{ N}$$

$$F_{Ax} = -1800 \text{ N}$$

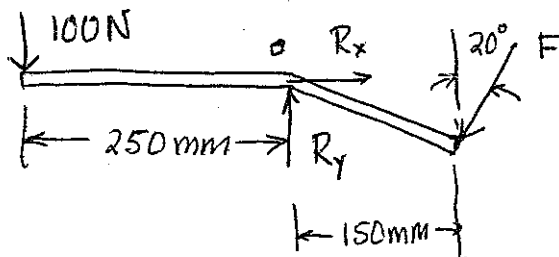
$$F_{By} = 1350 \text{ N}$$

$$F_{Ay} = -1350 \text{ N}$$



1-15

FBD OF THE LINK



$$\sum M_o = 0 \quad \uparrow \quad 100 \times 250 - F(150) \cos 20^\circ = 0$$
$$F = 177.4 \text{ N}$$

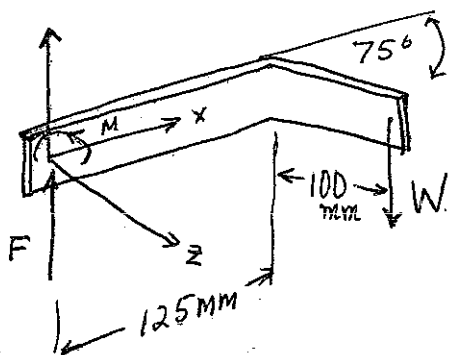
$$\sum F_x = 0 \quad R_x - F \sin 20^\circ = 0$$

$$R_x = 60.67 \text{ N}$$

$$\sum F_y = 0 \quad R_y - 100 - F \cos 20^\circ = 0$$

$$R_y = 266.7 \text{ N}$$

1.16



FBD OF BRACKET

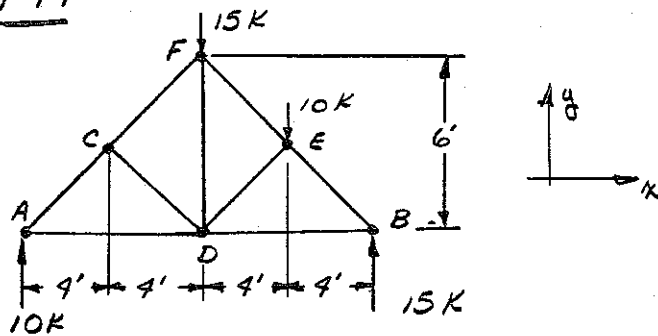
$$\sum M_3 = 0 \quad \uparrow$$

$$M_A - W(125 + 100 \cos 75^\circ) = 0$$

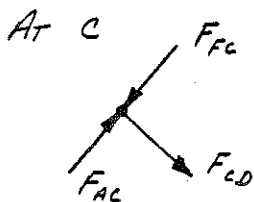
$$M_A = 100 \text{ N} \cdot \text{m}$$

$$W = 662 \text{ N}$$

1-17



FROM EXAMPLE 1-4 $F_{AC} = 16,670$ (C) C - COMPRESSION
 $F_{AD} = 13,330$ (T) T - TENSION



$$\sum F_x = 0$$

$$-F_{FC} \left(\frac{4}{5}\right) + F_{CD} \left(\frac{4}{5}\right) + F_{AC} \left(\frac{4}{5}\right) = 0$$

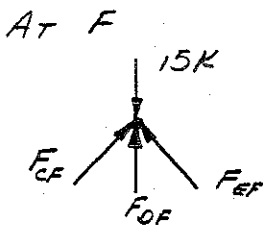
$$F_{CD} - F_{FC} = -16670$$

$$\sum F_y = 0$$

$$-F_{CD} \left(\frac{3}{5}\right) - F_{FC} \left(\frac{3}{5}\right) + F_{AC} \left(\frac{3}{5}\right) = 0$$

$$F_{CD} + F_{FC} = 16670$$

$$F_{CD} = 0 \quad \therefore F_{FC} = 16670$$



$$\sum F_x = 0$$

$$F_{CF} \left(\frac{4}{5}\right) - F_{EF} \left(\frac{4}{5}\right) = 0$$

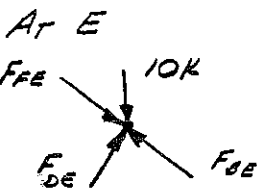
$$F_{EF} = F_{CF} = 16670$$

$$\sum F_y = 0$$

$$-15000 + F_{DF} + F_{CF} \left(\frac{3}{5}\right) + F_{EF} \left(\frac{3}{5}\right) = 0$$

$$F_{DF} = 15000 - 2 \left(\frac{3}{5}\right) (16670)$$

$$F_{DF} = -5000$$



$$\sum F_x = 0$$

$$F_{FE} \left(\frac{4}{5}\right) + F_{DE} \left(\frac{4}{5}\right) - F_{BE} \left(\frac{4}{5}\right) = 0$$

$$F_{DE} - F_{BE} = -F_{FE} = -16670$$

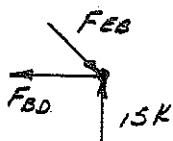
$$\sum F_y = 0$$

$$-10000 - F_{FE} \left(\frac{3}{5}\right) + F_{DE} \left(\frac{3}{5}\right) + F_{BE} \left(\frac{3}{5}\right) = 0$$

$$F_{DE} + F_{BE} = \frac{5}{3} (10,000) + 16670$$

$$F_{DE} = 8330$$

$A_T \quad B$



$$\sum F_x = 0$$

$$-F_{ED} + F_{EB} \left(\frac{4}{3} \right) = 0$$

$$F_{80} = 25000\left(\frac{4}{5}\right) = 20000 (T)$$

1.18 Equilibrium:

$$\sum F_x = 0 : F_{x6} + 500 - 400 = 0$$

$$\sum F_y = 0 : F_{YG} = 0$$

$$\sum F_z = 0: F_z G - 1000 = 0$$

$$\sum M_{G_x} = 0 : -1000 \times 2 + M_{xG} = 0$$

$$\sum M_{G_y} = 0 : 500 \times 25 - 400 \times 25 + M_{yG} = 0$$

$$\sum M_{G_2} = 0 : 400 \times 3 + 500 \times 3 + M_{G_2} = 0$$

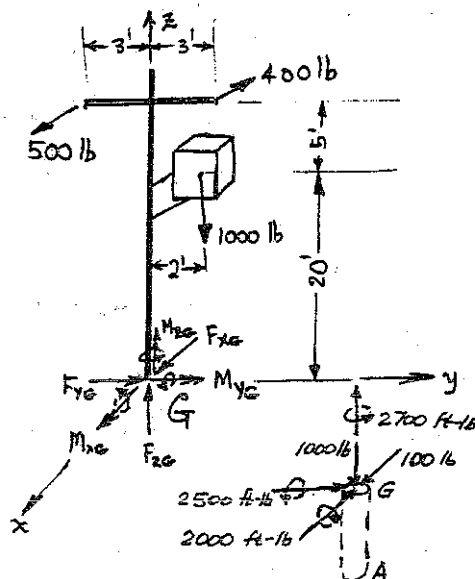
Lösung, $\vec{F}_G = (-100\vec{i} + 1000\vec{k})\text{ lb}$

Note: these are

five elements

acting on POLE

Note: these $\vec{F}$ are
force & moment
acting on POLE. $\vec{M}_G = (2000\vec{i} - 2500\vec{j} - 2700\vec{k}) \text{ lb}\cdot\text{ft}$



1.19 Idealisation: Assume thrust to act through hub of propeller. Neglect drag forces.

Free-body of engine + propeller:

$$\sum \bar{M} = 0 : M_{\text{CASING}} = M_{\text{AIR}} = 20,000 \text{ N}\cdot\text{m} \quad (F_{\text{STRET}})$$

$$\Sigma F = 0 : (F_{\text{STRUT}})_{\text{c}} = F_{\text{AIR}} = \text{THRUST} = 17.6 \text{ kN}$$

$$(F_{STRUT})_y = W = 11 \text{ kN}$$

Thus the propeller thrust and torque are transmitted without change by the engine to the strut.

Free-Body of Strut

Equilibrium: $\sum F_y = 0 : F_{Ay} = 11 \text{ kN}$

$$\sum F_x = 0: F_{Ax} = -17.6 \text{ kN}$$

$$\sum M_A = 0:$$

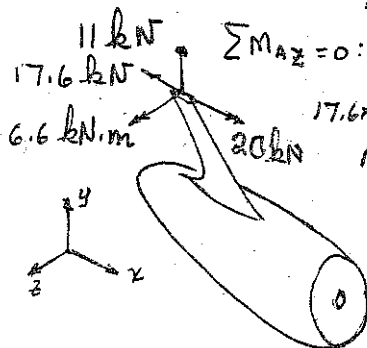
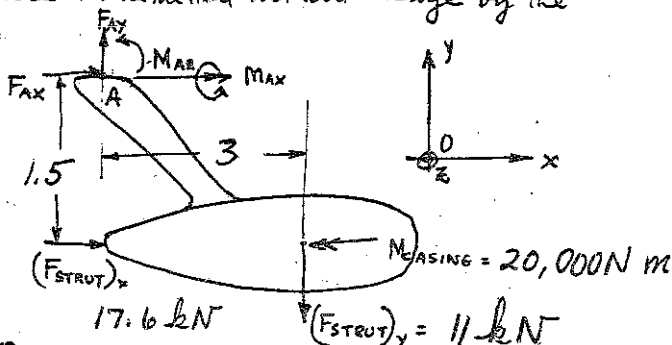
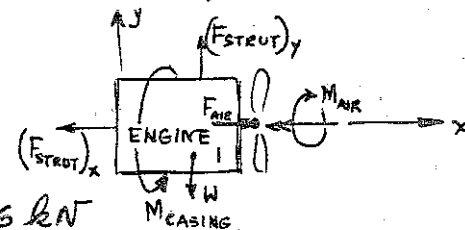
$$17.6 \times 1.5 - 11 \times 3 + M_{A2} = 0$$

$$M_{A2} = 66 \text{ kN}\cdot\text{m}$$

$$\Sigma M_{Ax} = 0 : M_{Ax} = M_{\text{Casing}} = 20 \text{ kN.m}$$

FORCE + MOMENT EXERTED ON WING

$$\vec{F} = (17.6 \hat{i} - 11 \hat{j}) \text{ kN}; \vec{M} = -(20 \hat{i} + 6.6 \hat{j}) \text{ kN}\cdot\text{m}$$



1.20 The procedure is to assume stability and then see if the necessary forces can exist.

Assumptions: Zero friction at ball surfaces; infinite friction between table and can.

Free-body of both balls together:

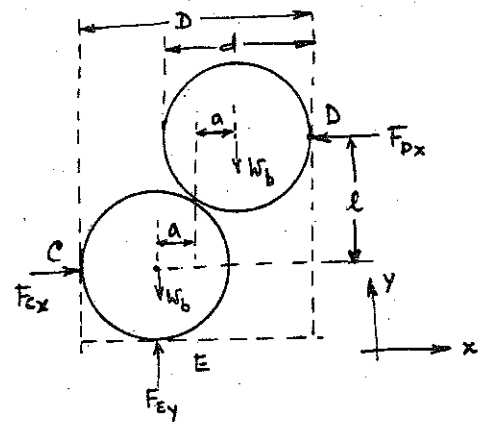
Geometry: $a = \frac{D-d}{2}$
 $l = 2\sqrt{\left(\frac{d}{2}\right)^2 - a^2} = \sqrt{D(2d-D)}$

Equilibrium: $\sum F_y = 0: F_{Ey} = 2W_b$

$\sum F_x = 0: F_{Ex} = F_{Dx}$

$\sum M_{CE} = 0: (F_{Dx})l + (F_{Ey})\frac{d}{2} = W_b\frac{d}{2} + W_b\left(\frac{d}{2} + 2a\right)$

or $F_{Ex} = F_{Dx} = \frac{W_b(D-d)}{l} = \frac{W_b(D-d)}{\sqrt{D(2d-D)}}$



Free-Body of Can: Suppose equilibrium is possible.

The force F_p is the reaction of the table, at some radius x from the axis.

Equilibrium: $\sum M_{B_z} = 0:$

$W_c \cdot \frac{D}{2} + F_{Ex} \cdot \frac{d}{2} = F_{Dx} \left(\frac{d}{2} + l\right) + F_p \left(\frac{D}{2} - x\right)$

$\sum F_y = 0: F_p = W_c$

These two equations give

$x = \frac{W_b}{W_c} (D-d)$

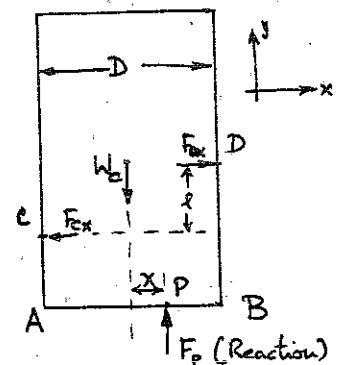
FOR OJ: $\frac{W_b}{W_c} = \frac{7}{2}; D = 50 \text{ mm}$

$x = \frac{2.0}{0.57} (50 - 45) = 17.5 < \frac{D}{2}$
 EQUILIBRIUM POSSIBLE

FOR BEER CAN $\frac{W_b}{W_c} = \frac{2.0}{1.0} = 2$
 $D = 70 \text{ mm}$

$x = 2 (70 - 45) = 50 \text{ mm}$

$x > \frac{D}{2}$ EQUILIBRIUM NOT POSSIBLE



- 1-21 Idealisation: (a) inelastic contact
(b) no friction anywhere
(c) no deformation of rod

Assume Equilibrium exists.

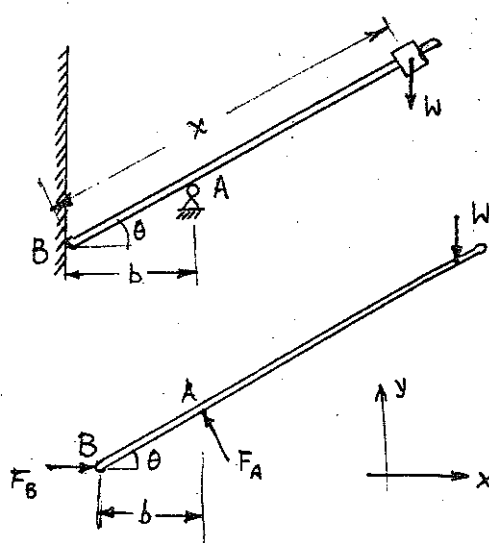
Equilibrium of free-body:

$$\sum F_x = 0: F_B = F_A \sin \theta$$

$$\sum F_y = 0: F_A \cos \theta = W$$

$$\sum M_{Bz} = 0: F_A \cdot \frac{b}{\cos \theta} = W \cdot x \cos \theta$$

Solving, $x = \frac{b}{\cos^3 \theta}$

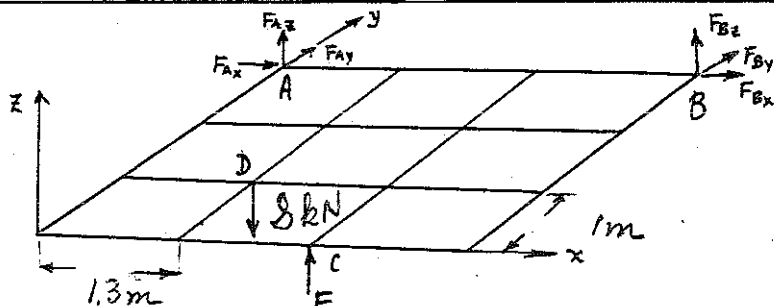


-22

Idealisation:

- (a) No friction at hinges
(b) Prop is rigid
(c) No friction at prop.

Equilibrium:



$$\sum F_x = 0: F_{Ax} + F_{Bx} = 0 \text{ These are indeterminate. Assume } F_{Ax} = F_{Bx} = 0.$$

$$\sum F_y = 0: F_{Ay} + F_{By} = 0$$

$$\sum F_z = 0: F_{Az} + F_{Bz} + F_C = 8 \text{ kN}$$

$$\sum M_{Bz} = 0: F_{Ay} = 0$$

$$\sum M_{Az} = 0: F_{By} = 0$$

$$\sum M_{Ay} = 0: F_{Bz} \times 3.9 + F_C \times 2.6 = 8 \times 1.3$$

$$\sum M_{Ax} = 0: F_C \times 3 = 8 \times 2$$

Solving,

$$F_{Az} = 3.55 \text{ kN}$$

$$F_{Bz} = -0.88 \text{ kN}$$

$$F_C = 5.33 \text{ kN}$$

1.23

Equilibrium:

$$\sum M_{Ox} = 0:$$

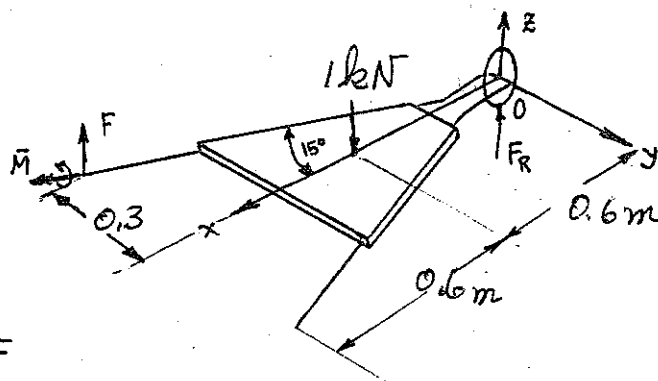
$$M \cos 15^\circ = F \times 0.3$$

$$\sum M_{Oy} = 0$$

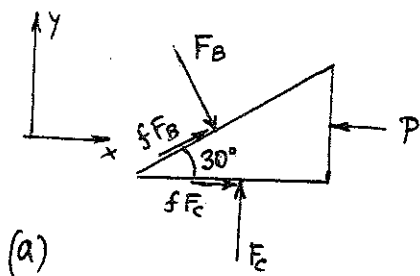
$$M \sin 15^\circ = 1 \times 0.6 - 1.2 F$$

Solve: $M = 0.145 \text{ kN} \cdot \text{m}$

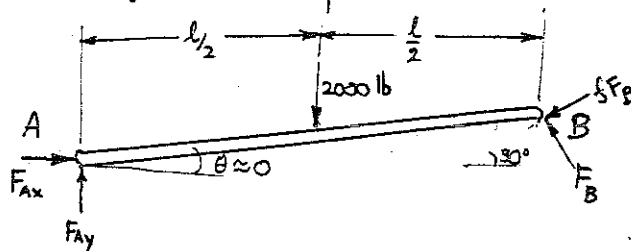
$$F = 0.468 \text{ kN}$$



1.24 Assuming a general friction coefficient f , when the wedge is moving, tangential force at contact = $f \times$ normal force.



(a)



Equilibrium of AB:

$$\sum M_A = 0: -2000 \times \frac{l}{2} + F_B \cos 30^\circ \times l - f F_B \sin 30^\circ \times l = 0$$

$$\therefore F_B = \frac{1000}{0.866 - \frac{1}{2}f}$$

Equilibrium of wedge:

$$\sum F_y = 0: F_c - F_B \cos 30^\circ + f F_B \sin 30^\circ = 0$$

$$\sum F_x = 0: -P + f F_c + (f \cos 30^\circ + \sin 30^\circ) F_B = 0$$

These give $F_c = 1000 \text{ lb}$.

$$P = 1000 \left[f + \frac{0.866f + 0.5}{0.866 - 0.5f} \right]$$

For $f = 0.3$, $P = 1364 \text{ lb}$

(b) If f is very small, the wedge will slip out when the force P is removed. The borderline occurs when the wedge is just prevented

1-24 CON'T

from slipping out by the friction forces. We go through the previous analysis, replacing f by $-f$ everywhere, because the tendency we are now investigating is slipping in the opposite direction.

Thus we get

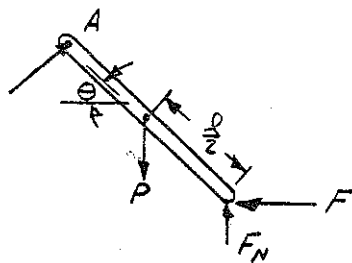
$$P = 1000 \left[-f + \frac{0.5 - 0.866f}{0.866 + 0.5f} \right]$$

We require $P = 0$, or

$$\frac{0.5 - 0.866f}{0.866 + 0.5f} - f = 0 \quad \text{This gives } \underline{f = 0.27} \text{ (borderline)}$$

1-25

FREE BODY DIAGRAM OF THE LINK (CCW ROTATION)



FRICTION FORCE

$$F = f F_N$$

$$\sum M_A = 0$$

$$-F(l) \sin \theta - P\left(\frac{l}{2}\right) \cos \theta + F_N(l) \cos \theta = 0$$

$$F\left(\frac{1}{f} - \tan \theta\right) = \frac{P}{2}$$

$$F = \frac{Pf}{2(1 - f \tan \theta)}$$

FOR CW ROTATION F IS IN THE OPPOSITE DIRECTION

$$\sum M_A = 0$$

$$F(l) \sin \theta - P\left(\frac{l}{2}\right) \cos \theta + F_N(l) \cos \theta = 0$$

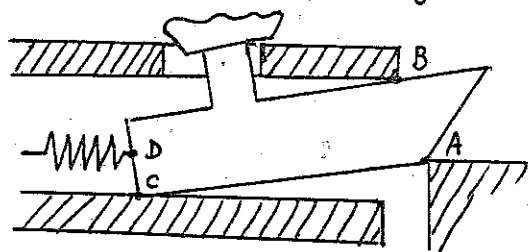
$$F\left(\frac{1}{f} + \tan \theta\right) = \frac{P}{2}$$

$$F = \frac{Pf}{2(1 + f \tan \theta)}$$

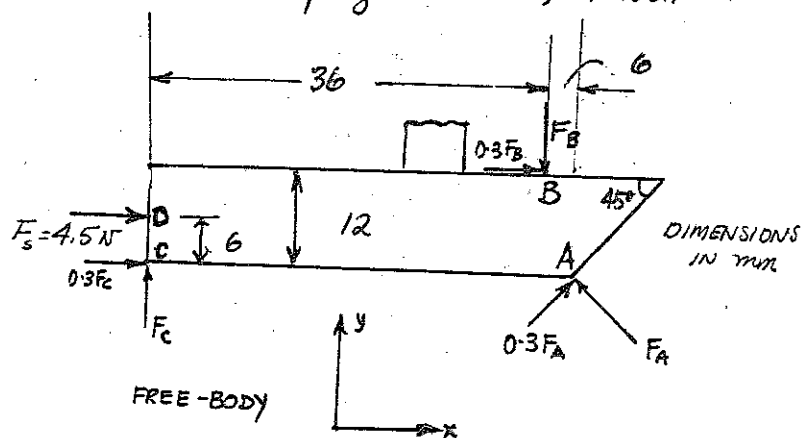
THE MECHANISM BECOMES A FRICTION LOCK FOR CCW-ROT.

$$F \rightarrow \infty \quad \text{FOR} \quad \tan \theta = \frac{1}{f}$$

1.26 Idealisation: all contacts are inelastic. As the body begins to slide, it will appear as shown in Fig (i)



(a) Fig (i)



Equilibrium of free-body:

$$\sum F_x = 0: F_s + 0.3(F_c + F_B) - \frac{1}{\sqrt{2}}(F_A - 0.3F_A) = 0 \quad \text{--- (i)}$$

$$\sum F_y = 0: F_c - F_B + \frac{1}{\sqrt{2}}(F_A + 0.3F_A) = 0 \quad \text{--- (ii)}$$

$$\sum M_A = 0: F_s \times 6 + F_c \times 42 + F_B \times 0.3 \times 12 - F_B \times 6 = 0 \quad \text{--- (iii)}$$

Solving, $F_A = 22.05 \text{ N}$; $F_B = 20.83 \text{ N}$; $F_c = 0.549 \text{ N}$

Free-body of door: $\sum M_H = 0$:

$$Pl + \frac{0.3F_A}{\sqrt{2}}e - \frac{F_A}{\sqrt{2}}e - \frac{F_A}{\sqrt{2}}(l+6) - \frac{0.3F_A}{\sqrt{2}}(l+6) = 0$$

$$\text{or } \left(P - \frac{1.3F_A}{\sqrt{2}}\right)l - \frac{0.7F_A}{\sqrt{2}}e - \frac{1.3F_A}{\sqrt{2}} \times 6 = 0.$$

Since $l \gg e$ and $l \gg 6$, neglect the last two terms and obtain $P = 20.26 \text{ N}$

(b) Without friction, equations (i) - (iii) may be modified, to become

$$F_s - \frac{F_A}{\sqrt{2}} = 0$$

$$F_c - F_B + \frac{F_A}{\sqrt{2}} = 0$$

$$6F_s + 0.42F_c - 6F_B = 0$$

Solving, $F_A = 6.363 \text{ N}$, $F_B = 4.5 \text{ N}$, $F_c = 0$

a free-body diagram of the door, as in part (a) gives

$$P = \frac{F_A}{\sqrt{2}} = \underline{\underline{4.5 \text{ N}}}$$

Idealisation: supports A, B, C transmit no torques or moments, are frictionless.

Preliminary: Assume mean tension is same after loading as before.

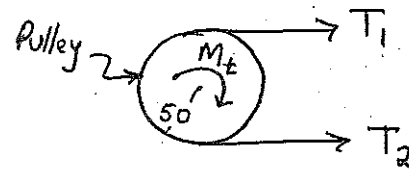
$$\frac{T_1 + T_2}{2} = T_m = 125 \text{ N}$$

EQUILIBRIUM OF PULLEY $\Sigma M_o = 0$

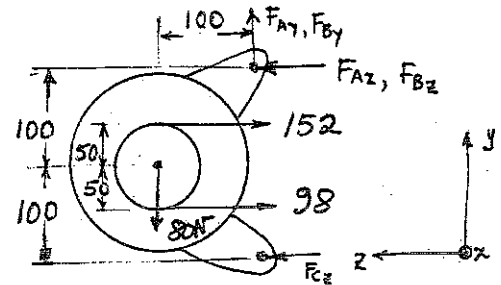
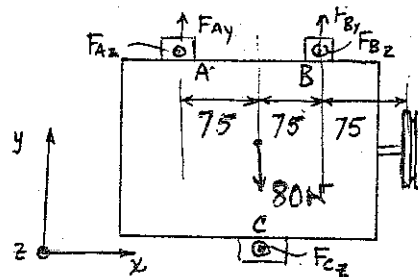
$$\therefore M_t = 2.7 \text{ N}\cdot\text{m} = T_2 \times 50' - T_1 \times 50$$

TO FIND $T_1 = 98 \text{ N}$

$$T_2 = 152 \text{ N}$$



M_t = TORQUE SUPPLIED TO
Pulley by MOTOR



EQUILIBRIUM $\Sigma F_x = 0$ SATISFIED

$$\Sigma F_y = 0: F_{Ay} + F_{By} - 80 = 0$$

$$\Sigma F_z = 0: F_{Az} + F_{Bz} + F_{Cz} - 152 - 98 = 0$$

$$\Sigma M_{Ax} = 0: 80 \times 100 + 152 \times 50 + 98 \times 150 = F_{Cz} \times 200$$

$$\Sigma M_{Ay} = 0: -F_{Bz} \times 150 + 152 \times 225 + 98 \times 225 - F_{Cz} \times 75 = 0$$

$$\Sigma M_{Az} = 0: -80 \times 75 + F_{By} \times 150 = 0$$

Solution

$$F_{Ay} = 40 \text{ N}$$

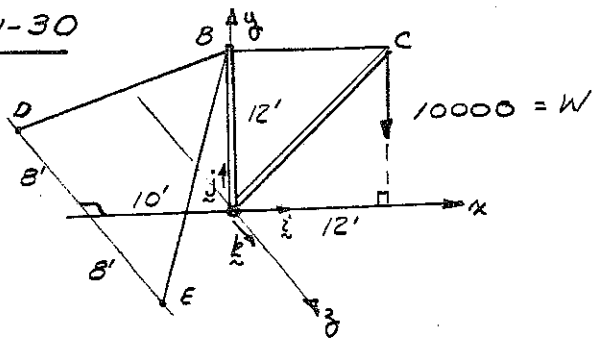
$$F_{Az} = -200.75 \text{ N}$$

$$F_{By} = 40 \text{ N}$$

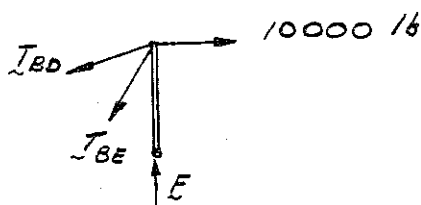
$$F_{Cz} = 151.5 \text{ N}$$

$$F_{Bz} = 299.25 \text{ N}$$

1-30



FREE BODY DIAGRAM



$$W = 10000 \hat{j}$$

$$\underline{T}_{BE} = T_{BE} (-10\hat{i} + 12\hat{j} + 8\hat{k}) / \sqrt{308}$$

$$\underline{T}_{BD} = T_{BD} (-10\hat{i} + 12\hat{j} - 8\hat{k}) / \sqrt{308}$$

$$\Sigma \underline{F} = 0$$

$$\underline{F} + \underline{W} + \underline{T}_{BE} + \underline{T}_{BD} = 0$$

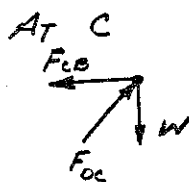
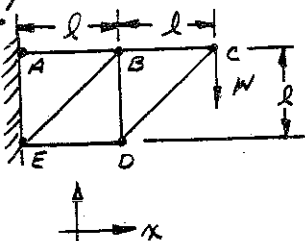
IN COMPONENT FORM

$$\hat{i} \quad -\frac{10}{\sqrt{308}} T_{BE} - \frac{10}{\sqrt{308}} T_{BD} + 10000 = 0$$

$$\hat{k} \quad \frac{8}{\sqrt{308}} T_{BE} - \frac{8}{\sqrt{308}} T_{BD} = 0$$

$$T_{BE} = T_{BD} = \frac{(10000) \sqrt{308}}{20} = 8,800 \text{ lb}$$

1-31



$$\Sigma F_x = 0$$

$$-F_{CB} + \frac{\sqrt{2}}{2} F_{DC} = 0$$

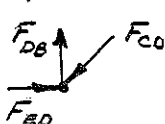
$$\Sigma F_y = 0$$

$$-W + \frac{\sqrt{2}}{2} F_{DC} = 0$$

$$F_{DC} = \sqrt{2} W$$

$$F_{CB} = W$$

AT D



$$\Sigma F_x = 0$$

$$F_{ED} - \frac{\sqrt{2}}{2} F_{CD} = 0$$

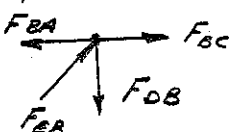
$$F_{ED} = W$$

$$\Sigma F_y = 0$$

$$F_{DB} - \frac{\sqrt{2}}{2} F_{CD} = 0$$

$$F_{DB} = W$$

AT B



$$\Sigma F_x = 0$$

$$F_{BC} + \frac{\sqrt{2}}{2} F_{EB} - F_{BA} = 0$$

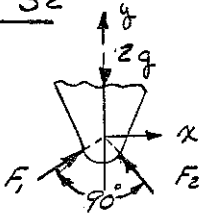
$$F_{BA} = 2W$$

$$\Sigma F_y = 0$$

$$-F_{DB} + \frac{\sqrt{2}}{2} F_{CB} = 0$$

$$F_{DB} = \sqrt{2} W$$

1-32



$$\sum F_x = 0$$

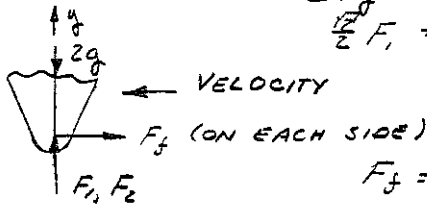
$$\frac{\sqrt{2}}{2} F_1 - \frac{\sqrt{2}}{2} F_2 = 0$$

$$F_1 = F_2$$

$$\sum F_y = 0$$

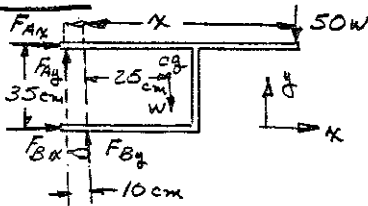
$$\frac{\sqrt{2}}{2} F_1 + \frac{\sqrt{2}}{2} F_2 - 2q = 0$$

$$F_1 = F_2 = \sqrt{2} q$$



$$F_f = \pm F_{1,2} = (.2)(\sqrt{2}q) = .282q$$

1-33



$$(1) \sum F_x = 0$$

$$F_{Ax} + F_{Bx} = 0$$

$$(2) \sum F_y = 0$$

$$-50W - W + F_{By} + F_{Ay} = 0$$

$$(3) \sum M_A = 0$$

$$-50W(x+10) - W(35) + F_{By}(10) + F_{Bx}(35) = 0$$

$$(4) F_{Ay} = \pm F_{Ax} \text{ AND } F_{By} = \pm F_{Bx} \text{ (FOR IMPENDING}$$

FROM (4), (1), (2)

$$F_{Ax} = \frac{51W}{2f} = \frac{51W}{2(.3)}$$

MOTION DOWNWARD
BOTH FORCES ARE AS
SHOWN)

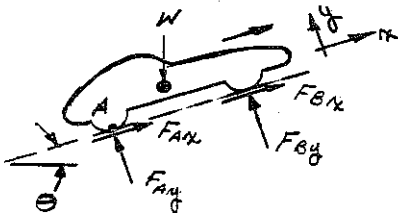
$$-F_{Ax} = F_{Bx}$$

FROM (3)

$$-50W(x+10) - W(35) + \frac{(.3)(51W)}{2(.3)}(10) - \frac{51W}{.6}(35) = 0$$

$$x = 53 \text{ cm}$$

1-34



$$\sum F_x = 0$$

$$F_{Ax} + F_{Bx} - W \sin \theta = 0$$

$$\sum F_y = 0$$

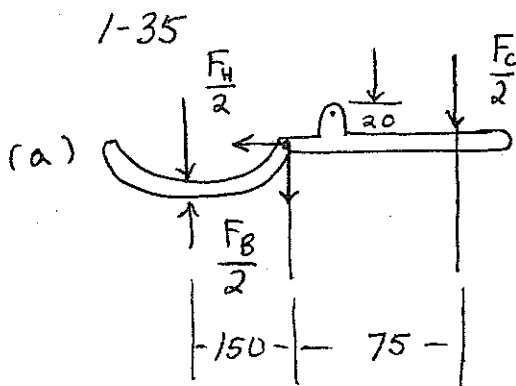
$$F_{Ay} + F_{By} - W \cos \theta = 0$$

FROM FRICTION RELATION

$$F_{Ax} = \pm F_{Ay} \text{ \& } F_{Bx} = \pm F_{By}$$

$$\frac{W \sin \theta}{W \cos \theta} = \frac{F_{Ax} + F_{Bx}}{F_{Ay} + F_{By}}$$

$$\tan \theta = .7 \text{ AND } \theta = 35^\circ$$

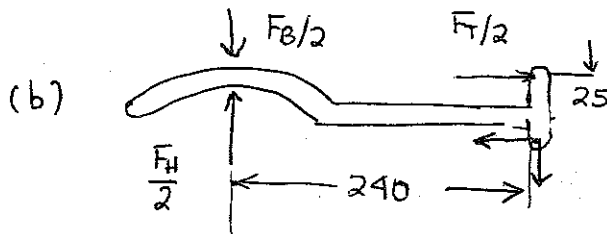


$$\sum M_P = 0 \rightarrow$$

$$-\frac{F_B}{2}(150) + \frac{F_H}{2}(150) - \frac{F_T}{2}(20) - \frac{F_C}{2}(75) = 0$$

$$F_H = F_B + \frac{1}{150} [120 \times 20 + 20 \times 75]$$

$$F_H = F_B + 26 \text{ N}$$



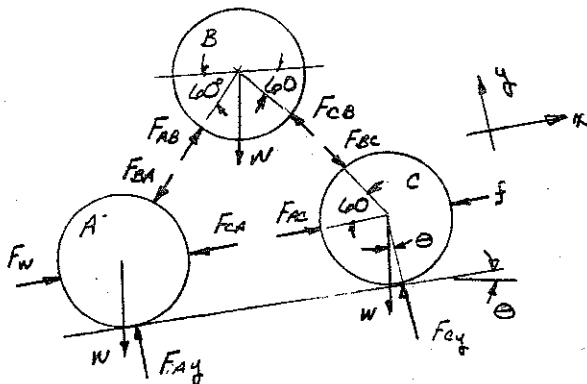
$$\sum M_P = 0 \rightarrow$$

$$\frac{F_B}{2}(240) - \frac{F_H}{2}(240) - \frac{F_T}{2}(25) = 0$$

$$F_H = F_B - \frac{1}{240} \times 120 = F_B - 0.5 \text{ N}$$

DESIGN (b) IS BETTER IT ACTS IN A MANNER WHICH RELIEVES THE FORCE OF THE BIRTH CANAL ON THE FETAL HEAD.

1-36



ASSUME THE LOGS ARE ALL OF THE SAME DIAMETER

WHEN C IS ABOUT TO MOVE
 $f = 0$ AND $F_{AC} = 0$

FOR C, $\sum F_x = 0$

$$F_{BC} \cos 60 - W \sin \theta = 0$$

$$\sin \theta = \frac{F_{BC} \cos 60}{W}$$

FOR B $\sum F_x = 0$ $-F_{CB} \cos 60 + F_{AB} \cos 60 - W \sin \theta = 0$

$\sum F_y = 0$ $F_{CB} \sin 60 + F_{AB} \sin 60 - W \cos \theta = 0$

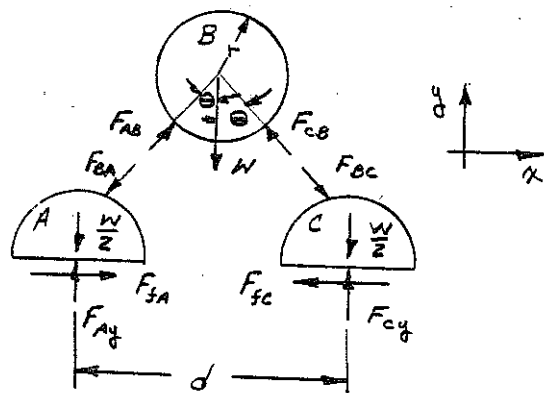
$$\text{SOLVING FOR } F_{CB} = F_{BC} = \frac{W (\sin \theta \sin 60 - \cos \theta \cos 60)}{-2 \sin 60 \cos 60}$$

$$= \frac{W \cos (\theta + 60)}{\cos 30}$$

$$\sin \theta = \frac{W \cos (\theta + 60) \cos 60}{\cos 30}$$

$$\tan \theta = \frac{2}{3} \frac{\cos^2 60}{\cos 30} \quad \theta = \tan^{-1} \frac{\sqrt{3}}{9} = 10.8^\circ$$

1-37



FOR EQUILIBRIUM OF B

$$\sum F_x = 0$$

$$(1) F_{AB} \sin \theta - F_{CB} \sin \theta = 0$$

$$F_{AB} = F_{CB}$$

$$\sum F_y = 0$$

$$(2) F_{AB} \cos \theta + F_{CB} \cos \theta - W = 0$$

FROM GEOMETRY

$$\frac{d}{2} = 2r \sin \theta$$

FOR A

$$(3) \sum F_x = 0 \quad -F_{BA} \sin \theta + F_{fA} = 0$$

$$(4) \sum F_y = 0 \quad -F_{BA} \cos \theta + F_{Ay} - \frac{W}{2} = 0$$

FROM FRICTION CONSIDERATIONS $F_{fA} = f F_{Ay}$ (5)
 $F_{AB} = F_{BA}$ AS SHOWN

FROM (1) & (2)

$$F_{AB} \cos \theta = \frac{W}{2}$$

WITH (4) & (5)

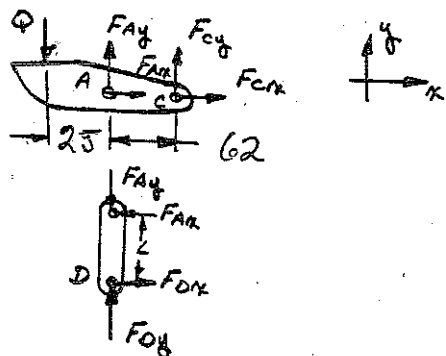
$$F_{Ay} = W \quad \& \quad F_{fA} = f W$$

$$\frac{F_{AB} \sin \theta}{F_{AB} \cos \theta} = \tan \theta = \frac{f W}{\frac{W}{2}}$$

$$\theta = \tan^{-1}(2f)$$

$$d = 2r \sin(\tan^{-1} 2f)$$

1-38



$$\sum M_A = 0$$

$$F_{Cy} (62) - Q (25) = 0$$

$$\sum M_D = 0$$

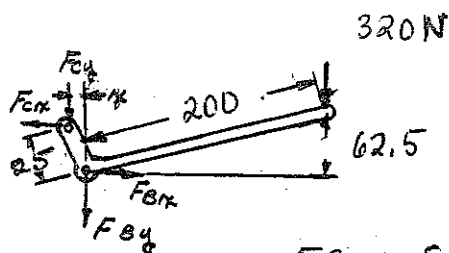
$$F_{Ax} (L) = 0$$

$$\sum F_x = 0$$

$$F_{Ax} + F_{Cx} = 0$$

$$F_{Ax} = F_{Cx} = 0$$

1-24



$$F_{Cy}(x) - 320 N (200^2 - 62.5^2)^{1/2} = 0$$

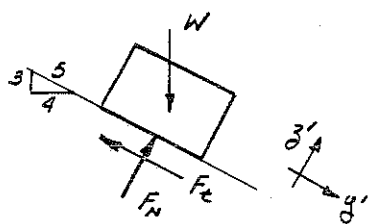
FROM SIMILAR TRIANGLES $\frac{x}{25} = \frac{62.5}{200}$

$$x = 7.81$$

$$F_{Cy} = 7784 N$$

$$\therefore Q = 19.3 \text{ kN}$$

1-39



$$\theta = \tan^{-1} \frac{3}{4}$$

$$\sum F_{y'} = 0$$

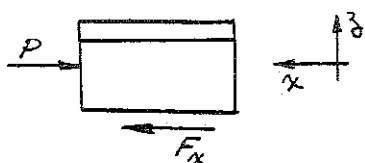
$$-F_t + W \sin \theta = 0$$

$$F_t = W \sin \theta = \frac{3}{5} W$$

$$\sum F_{x'} = 0$$

$$F_N - W \cos \theta = 0$$

$$F_N = W \cos \theta = \frac{4}{5} W$$



$$\sum F_x = 0$$

$$F_k - P = 0$$

$$F_k = P$$

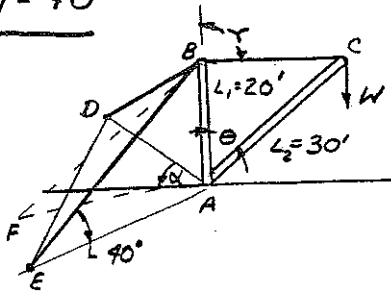
WHEN THE BLOCK BEGINS TO MOVE, $P = .4W$ AND THE MAGNITUDE OF THE FRICTION FORCE IS EQUAL TO THE MAGNITUDE OF THE VECTOR SUM OF $\vec{F}_t$ AND $\vec{F}_k$.

$$F_f = [F_t^2 + F_k^2]^{1/2}$$

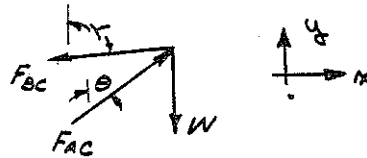
$$f F_N = [(.6W)^2 + (.4W)^2]^{1/2}$$

$$f = \frac{W (.52)^{1/2}}{.8W} = .9$$

1-40



IN PLANE ACBF



$$\sum F_x = 0$$

$$F_{AC} \sin \theta - F_{BC} \sin \gamma = 0$$

$$\sum F_y = 0$$

$$-W + F_{AC} \cos \theta - F_{BC} \cos \gamma = 0$$

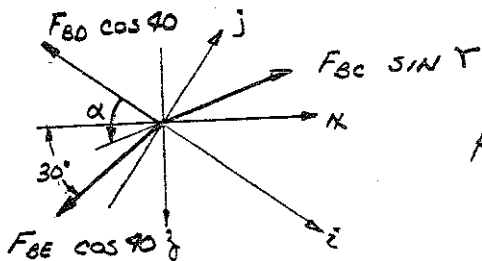
SOLVING $F_{BC} = \frac{W \sin \theta}{\sin (\gamma - \theta)}$

FROM GEOMETRY

$$\frac{L_2}{\sin (\pi - \gamma)} = \frac{L_1}{\sin (\gamma - \theta)}$$

$$\sin \gamma = \frac{L_2}{L_1} \sin (\gamma - \theta)$$

IN THE PLANE DAEF



$$\sum F_j = 0$$

$$F_{BC} \sin \gamma \cos (90 - \alpha) - F_{BE} \cos 40 \cos 30 = 0$$

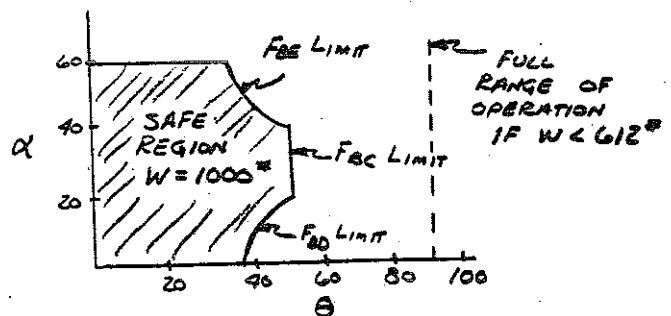
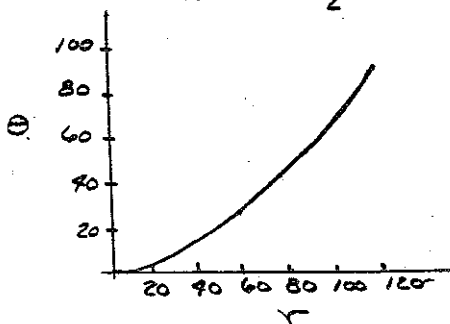
$$F_{BE} = W \left(\frac{L_2}{L_1} \right) \frac{\sin \theta \sin \alpha}{\cos 40 \cos 30}$$

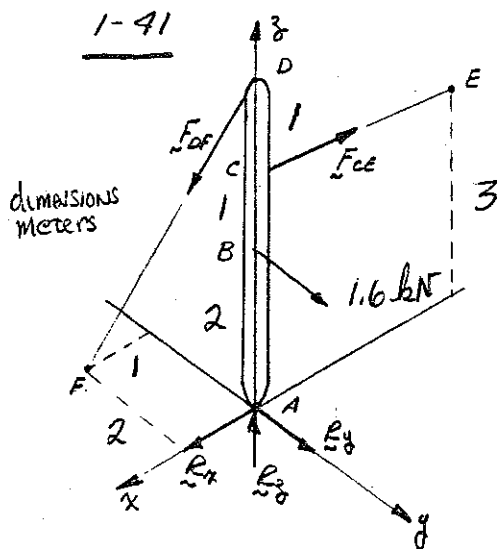
$$\sum F_i = 0$$

$$-F_{BD} \cos 40 + F_{BC} \sin \gamma \sin (90 - \alpha) - F_{BE} \cos 40 \sin 30 = 0$$

$$F_{BD} = W \frac{L_2}{L_1} \frac{\sin \theta \cos (\alpha + 30)}{\cos 30 \cos 40}$$

$$\sin \gamma = \frac{3}{2} \sin (\gamma - \theta)$$





$$\underline{F}_{DF} = F_{DF} (\underline{i} - 2\underline{j} - 4\underline{k}) / \sqrt{21}$$

$$\underline{F}_{CE} = -F_{CE} \underline{i}$$

$$\underline{R} = R_x \underline{i} + R_y \underline{j} + R_z \underline{k}$$

$$\text{APPLIED FORCE } \underline{F} \underline{j}$$

$$\Sigma \underline{F} = 0$$

$$\underline{i}: \frac{F_{DF}}{\sqrt{21}} - F_{CE} + R_x = 0$$

$$\underline{j}: -\frac{2F_{DF}}{\sqrt{21}} + F + R_y = 0$$

$$\underline{k}: -\frac{4F_{DF}}{\sqrt{21}} + R_z = 0$$

$$\Sigma \underline{M}_A = 0$$

$$\underline{M}_{DF} = \underline{r}_A \times \underline{F}_{DF} = \frac{1}{\sqrt{21}} (4\underline{j} + 8\underline{k}) F_{DF}$$

$$\underline{M}_{CE} = \underline{r}'_A \times \underline{F}_{CE} = -3F_{CE} \underline{j}$$

$$\underline{M}_{\text{APPLIED}} = \underline{r}''_A \times \underline{F}_{\text{APP}} = -2F \underline{i}$$

$$\underline{i}: \frac{8}{\sqrt{21}} F_{DF} - 2F = 0$$

$$\underline{j}: \frac{4}{\sqrt{21}} F_{DF} - 3F_{CE} = 0 \quad F = 1.6 \text{ kN}$$

$$\text{SOLVING } F_{DF} = 1.833 \text{ kN}$$

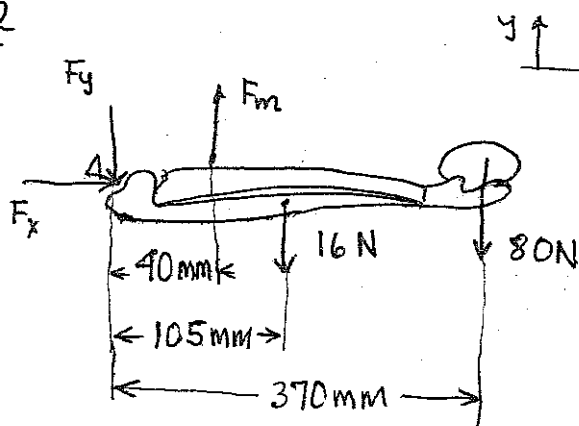
$$F_{CE} = 0.533 \text{ kN}$$

$$R_x = 0.133 \text{ kN}$$

$$R_y = 0.8 \text{ kN}$$

$$R_z = 1.60 \text{ kN}$$

1-42



$$(a) \quad \sum F_x = 0 \quad F_x = 0$$

$$\sum F_y = 0$$

$$-F_y + F_m - 16 - 80 = 0$$

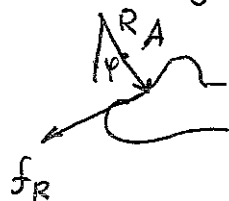
$$\sum M_A = 0 \quad \curvearrowright$$

$$F_m \times 40 - 16 \times 105 - 80 \times 370 = 0$$

$$F_m = 782 \text{ N}$$

$$F_y = 686 \text{ N}$$

(b) AT THE JOINT WITH FRICTION



$$\sum F_x = 0 \quad \tan \varphi = \mp f$$

$$\sum F_y = 0$$

$$-R \cos \varphi \pm f R \sin \varphi + F_m - 16 \text{ N} - 80 \text{ N} = 0$$

$$R \cos \varphi (1 \mp f \tan \varphi) = F_m - 96 = 686 \text{ N}$$

$$R \cos \varphi (1 + f^2) = 686 \text{ N}$$

$$\sum M_A = 0 \quad \curvearrowright :$$

$$\pm f R (20) + F_m (40) - 16 \times 105 - 80 \times 370 = 0$$

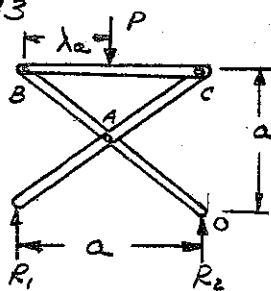
$$40 F_m \left[1 \pm \frac{0.5 f}{\cos \varphi (1 + f^2)} \right] = 31280 \mp \frac{f(20)(24)}{\cos \varphi (1 + f^2)}$$

$$\varphi = \tan^{-1}(\mp 0.015)$$

$$\approx 0.86^\circ$$

$$F_m = 782 \text{ N}$$

1-43



$$+\circlearrowleft \sum M_O = 0$$

$$P(a - \lambda a) - R_1 a = 0$$

$$R_1 = P(1 - \frac{\lambda a}{a})$$

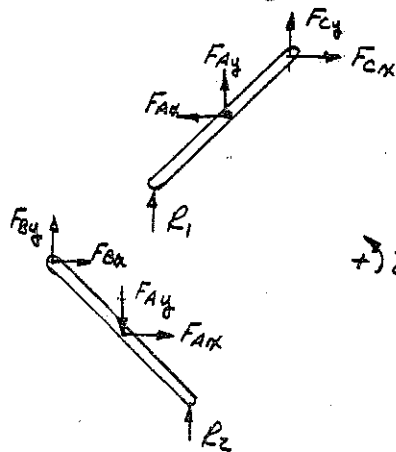
$$\sum F_y = 0$$

$$R_1 + R_2 - P = 0$$

$$R_2 = P - R_1 = \frac{\lambda a}{a} P$$

$$+\circlearrowleft \sum M_C = 0$$

$$-R_1 a - F_{Ax} \frac{a}{2} - F_{Ay} \frac{a}{2} = 0$$



$$+\circlearrowleft \sum M_B = 0$$

$$+ R_2 a + F_{Ax} \frac{a}{2} - F_{Ay} \frac{a}{2} = 0$$

$$F_{Ax} = -P$$

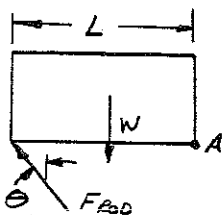
$$F_{Ay} = -P(1 - \frac{\lambda a}{a}) + P \frac{\lambda a}{a}$$

$$= -P(1 - \frac{2\lambda a}{a})$$

$$\text{MAGNITUDE OF SHEAR} = (F_{Ax}^2 + F_{Ay}^2)^{1/2} = P [1 + (1 - \frac{2\lambda a}{a})^2]^{1/2}$$

THIS IS A MAXIMUM OF $\sqrt{2} P$ WHEN $\lambda a = 0, a$

1-44



$$+\circlearrowleft \sum M_A = 0$$

$$W \frac{L}{2} - F_{Rod} L \cos \theta = 0$$

$$F_{Rod} L \cos \theta = \frac{WL}{2} = C \text{ (CONST.)}$$

ASSUME COST IS PROPORTIONAL TO THE VOLUME OF THE ROD.

$$\text{COST} \propto \text{AREA} \cdot \text{LENGTH} = A \frac{L}{\sin \theta}$$

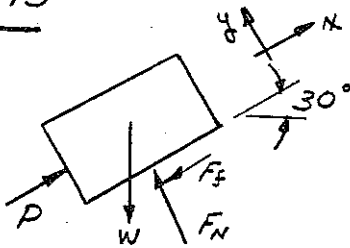
$$\propto \frac{C}{A}$$

$$\propto \frac{F_{Rod} \sin \theta \cos \theta}{2AC}$$

$$F_{Rod} \sin 2\theta$$

THE COST WILL BE A MINIMUM WHEN THE DENOMINATOR IS A MAXIMUM WRT θ . $\sin 2\theta = 1$ OR $\theta = 45^\circ$

1-45



$$\sum F_x = 0$$

$$P - F_f - W \sin 30 = 0$$

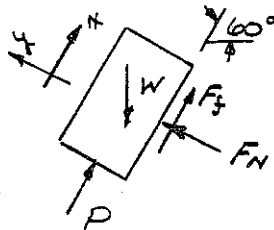
$$\sum F_y = 0$$

$$F_N - W \cos 30 = 0$$

$$F_N = W \cos 30$$

$$F_f = \mu F_N = \mu W \cos 30$$

$$P = \left(\mu \frac{\sqrt{3}}{2} + \frac{1}{2} \right) W$$



$$\sum F_x = 0$$

$$P + F_f - W \sin 60 = 0$$

$$\sum F_y = 0$$

$$F_N - W \cos 60 = 0$$

$$F_N = W \cos 60$$

$$F_f = \mu F_N = \mu W \cos 60$$

$$P = \left(-\mu \frac{1}{2} + \frac{\sqrt{3}}{2} \right) W$$

Assume the force required is the same for both cases.

$$\left(-\mu \frac{1}{2} + \frac{\sqrt{3}}{2} \right) W = \left(\mu \frac{\sqrt{3}}{2} + \frac{1}{2} \right) W$$

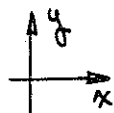
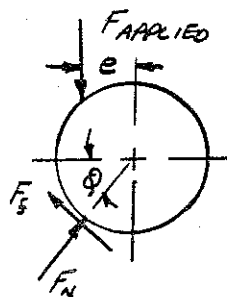
$$\mu = .27$$

1-96

THE PIN MAY BE CONSIDERED TO BE FREE OR WELDED INTO THE GRIP.

THE PIN WILL BE HELD BY FRICTION TO ONE SIDE.
THE FREE BODY DIAGRAM MUST BE DRAWN.

FOR A FREE PIN



$$\Sigma F_x = 0$$

$$F_N \cos \phi - F_f \sin \phi = 0$$

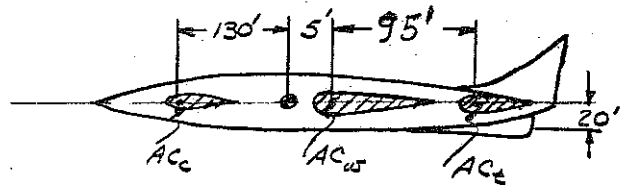
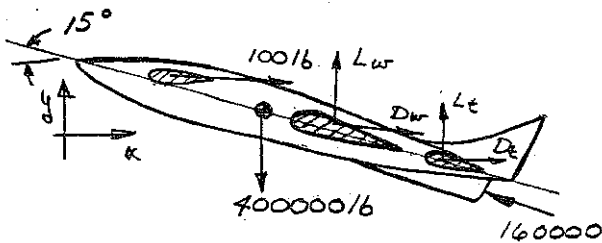
$$F_f = f F_N$$

$$\tan \phi = \frac{1}{f} = \frac{1}{.15} = 6.6$$

$$\phi = 81.5^\circ$$

$$\begin{aligned} e &= r_p \cos 81.5^\circ \\ &= (.125 \text{ in}) \cos 81.5^\circ \\ &= .018 \text{ in} \end{aligned}$$

1-47



$$+\sum M_{AC_w} = 0$$

$$- 100 \text{ lb} (135 \text{ ft} \sin 15^\circ) - (1.6 \times 10^5 \text{ lb}) (20 \text{ ft}) + (4.0 \times 10^5 \text{ lb}) (5 \text{ ft} \cos 15^\circ) + L_t (95 \text{ ft} \cos 15^\circ) + D_t (75 \text{ ft} \sin 15^\circ) = 0$$

$$\frac{L_t}{D_t} = 1.2 \quad \text{OR} \quad D_t = \frac{L_t}{1.2}$$

$$75 L_t \left(\cos 15^\circ + \frac{\sin 15^\circ}{1.2} \right) = (32 \times 10^5) - (20 \times 10^5 \cos 15^\circ) + (.14 \times 10^5 \sin 15^\circ)$$

$$L_t = 1.132 \times 10^4 \text{ lb}$$

$$\sum F_x = 0 \quad 100 + D_w + D_t - 1.6 \times 10^5 \cos 15^\circ = 0$$

$$D_w = 1.6 \times 10^5 \cos 15^\circ - \frac{1.43 \times 10^4}{1.2} - 100$$

$$= 1.639 \times 10^5 \text{ lb}$$

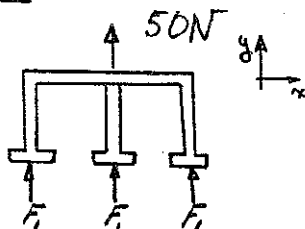
$$\sum F_y = 0 \quad - 4.0 \times 10^5 + L_w + L_t + 1.6 \times 10^5 \sin 15^\circ = 0$$

$$L_w = 4.0 \times 10^5 - 1.43 \times 10^4 - 1.6 \times 10^5 \sin 15^\circ$$

$$= 3.47 \times 10^5$$

$$\frac{L_w}{D_w} = \frac{3.47 \times 10^5}{1.639 \times 10^5} = 2.12$$

1-48



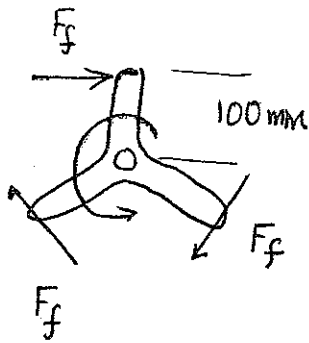
$$\sum F_y = 0$$

$$+ 3F_1 + 50 = 0$$

$$F_1 = -16.67 \text{ N TO HOLD AGAINST DRILL PRESSURE}$$

1-32

1.48 CONT



$$\sum M_o \curvearrowright$$

$$-3 F_f (0.1) + 15 \text{ N} \cdot \text{m} = 0$$

$$F_f = f F_2 = 0.4 F_2$$

$$F_2 = 125 \text{ N} \quad \text{TO HOLD AGAINST THE MOMENT}$$